CEG5101 Modern Computer Networking

Graduate Assistants: Ms. Poorvi Joshi (e1144005@u.nus.edu), Ms. Le Minh Chau (le_minhchau@u.nus.edu), Ms. Theepana Govintharajah (theepana@u.nus.edu)
Prepared by: Mr. Vijay Narasimha
Lecturer: Assoc. Prof. Mohan Gurusamy
ECE Department, CDE., National University of Singapore

Laboratory 04

Ryu Controller as a Learning Switch

LEARNING OBJECTIVES

The Learning Switch is a fundamental SDN application that enables switches to learn MAC addresses and make intelligent forwarding decisions, avoiding unnecessary packet flooding.

The main features of learning switch are:

1. MAC Address Learning
   - Automatically learns MAC addresses from incoming packets
   - Maps MAC addresses to specific switch ports
   - Maintains separate tables for each switch (by DPID)

```python
def __init__(self, *args, **kwargs):
    super(LearningSwitch13, self).__init__(*args, **kwargs)
    # MAC address table: MAC -> (switch_id, port)
    self.mac_to_port = {}
```

```python
# Learn the source MAC address
self.mac_to_port.setdefault(dpid, {})
self.mac_to_port[dpid][src] = in_port
self.logger.info("Learned MAC: %s -> switch %s, port %s", src, dpid, in_port)
```

2. Intelligent Forwarding
   - Known destinations: Forwards packets directly to learned ports
   - Unknown destinations: Floods packets to all ports (except incoming port)
   - Flow Rule Installation: Installs flow rules to handle future packets without controller intervention

```python
if dst in self.mac_to_port[dpid]:
    # Destination is known - forward to specific port
    out_port = self.mac_to_port[dpid][dst]
    self.logger.info("Forwarding to known host: port %s", out_port)
else:
    # Destination unknown - flood the packet
    out_port = ofproto.OFPP_FLOOD
    self.logger.info("Flooding packet")
```

DISCUSSION

Note: All files required for the exercises are under the provide **ceg_code** folder.

Task 1: Invoke the Ryu controller as a learning switch

1. In this activity, we will start Ryu as a learning switch.

2. We will start by issuing the command to bring up the Ryu controller as a learning switch "`ryu-manager learning_switch.py`".

3. We will then create a topology in mininet by issuing the command "`sudo mn --switch ovsk --controller remote --custom ceg_topo.py --topo cegTopo`".

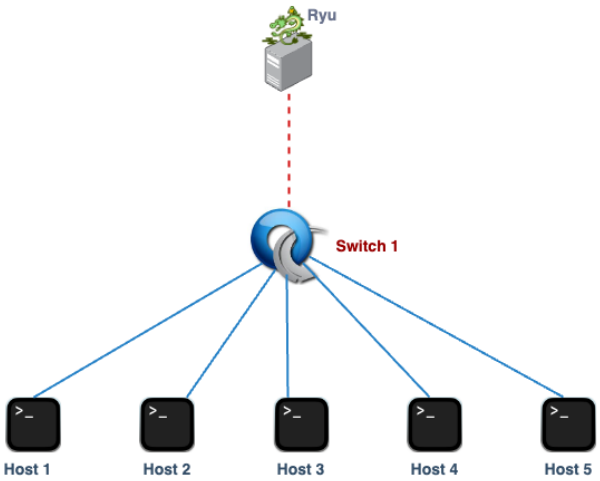
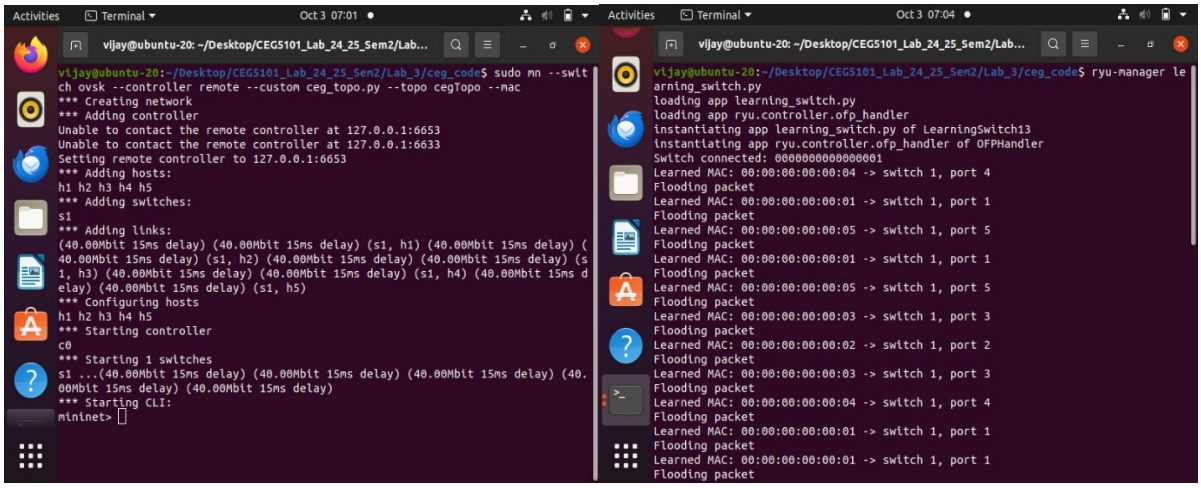
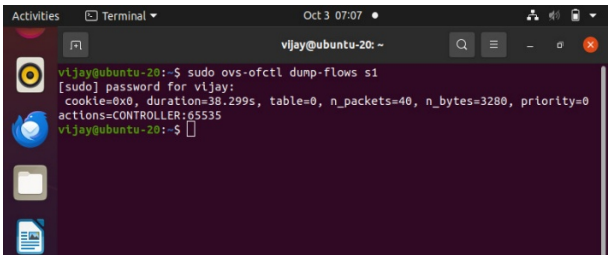


Figure 1: Topology

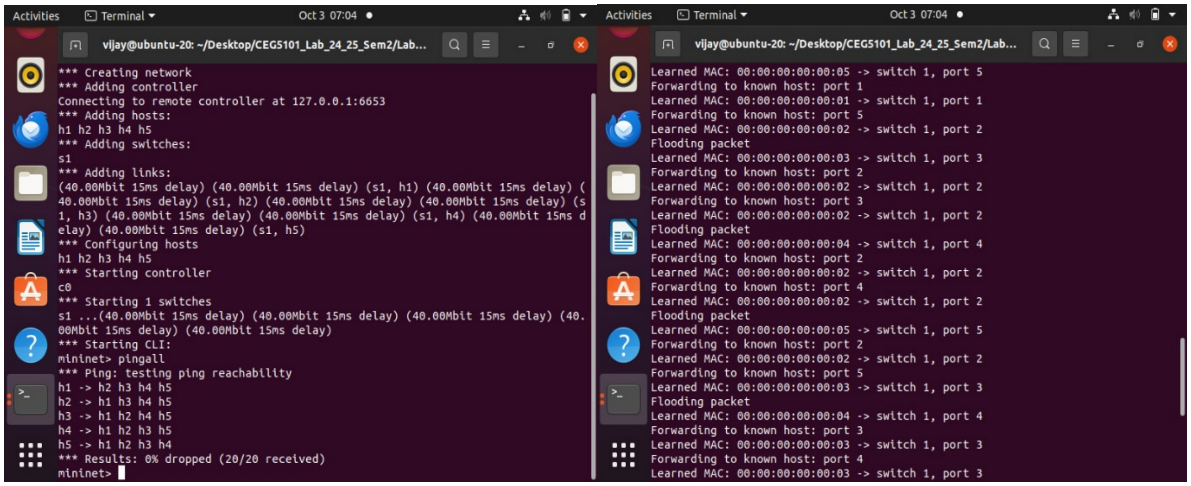


In the controller terminal, we see ARP broadcasting. Our flow rules only match packets that have the exact destination MAC address **and** arrive on the exact same input port. If a packet arrives on a different port (even with the same destination MAC) it would not match the existing flow and would be sent to the controller.

4. We will issue the command "`sudo ovs-ofctl dump-flows s1`". We observe that only the flow entry for the controller is present.



5. In Mininet CLI, we run "`pingall`". In the controller terminal, we observe that now packets are being forwarded directly to learned ports.



6. We will issue the command "`sudo ovs-ofctl dump-flows s1`". Along with the flow entry for the controller, there are now entries that map all ports to each other.

